# DFO, GDPopt and MINLP completion times — all nine models

Each cell: completion time / target attainment count. All 45 GDPopt and MINLP jobs are included.

| Model | DSDA | RBFOpt | BAM 5.0.0 | FMQA-SA | RIC GDPopt | GLOA GDPopt | LOA GDPopt | Big-M MINLP | Hull MINLP |
|---|---|---|---|---|---|---|---|---|---|
| CSTR 5<br>DFO B=8 | 21.72 s<br>3/10 | 12.73 s<br>10/10 | 24.80 s<br>10/10 | 21.92 s<br>3/10 | 31.27 s<br>1/1 | 3.76 s †<br>0/1 | 31.82 s<br>1/1 | 9.12 s<br>1/1 | 14.86 s †<br>1/1 |
| CSTR 10<br>DFO B=15 | 12.52 s<br>3/10 | 18.60 s<br>7/10 | 11.83 s<br>1/10 | 19.38 s<br>2/10 | 214.06 s<br>1/1 | 4.56 s †<br>0/1 | 234.13 s<br>1/1 | 123.74 s<br>1/1 | 126.47 s †<br>1/1 |
| CSTR 15<br>DFO B=22 | 29.21 s<br>2/10 | 48.99 s<br>10/10 | 19.50 s<br>1/10 | 39.55 s<br>1/10 | 1798.83 s TL<br>0/1 | 26.19 s †<br>0/1 | 1799.38 s TL<br>1/1 | 1799.00 s TL<br>1/1 | 940.36 s †<br>1/1 |
| CSTR 20<br>DFO B=29 | 27.47 s<br>3/10 | 53.70 s<br>9/10 | 31.50 s<br>2/10 | 55.76 s<br>1/10 | 1799.04 s TL<br>0/1 | 6.79 s †<br>0/1 | 1797.72 s TL<br>0/1 | 1798.72 s TL<br>1/1 | 1797.44 s TL<br>1/1 |
| CSTR 25<br>DFO B=36 | 46.30 s<br>3/10 | 99.81 s<br>10/10 | 46.12 s<br>2/10 | 67.17 s<br>1/10 | 1797.05 s TL<br>0/1 | 85.20 s †<br>0/1 | 1797.46 s TL<br>0/1 | 1797.67 s TL<br>1/1 | 1797.41 s TL<br>1/1 |
| CSTR 30<br>DFO B=43 | 51.13 s<br>5/10 | 97.91 s<br>10/10 | 33.87 s<br>2/10 | 74.05 s<br>1/10 | 1792.52 s TL<br>0/1 | 718.80 s †<br>0/1 | 1798.92 s TL<br>0/1 | 1797.31 s TL<br>1/1 | 1466.60 s †<br>0/1 |
| Column<br>DFO B=16 | 46.77 s<br>8/10 | 41.37 s<br>5/10 | 41.74 s<br>0/10 | 36.60 s<br>5/10 | 1797.54 s TL<br>0/1 | 1798.15 s TL<br>0/1 | 1797.77 s TL<br>0/1 | 34.36 s †<br>0/1 | 1797.17 s TL<br>0/1 |
| Small Batch (27 points)<br>DFO B=10 | 3.61 s<br>8/10 | 6.39 s<br>5/10 | 4.31 s<br>2/10 | 8.77 s<br>6/10 | 12.86 s<br>1/1 | 14.49 s<br>1/1 | 5.11 s<br>1/1 | 0.55 s<br>1/1 | 0.57 s<br>1/1 |
| Dynamic<br>DFO B=50 | 24.47 s<br>10/10 | 316.63 s<br>10/10 | 37.91 s<br>10/10 | 62.77 s<br>6/10 | 1797.71 s TL<br>0/1 | 1797.55 s TL<br>0/1 | 1797.84 s TL<br>0/1 | 1797.51 s TL<br>0/1 | 1797.77 s TL<br>0/1 |